Vertically Integrated Industries

Initial takeaways on minerals, metals, manufacturing and chemistry

1. Tailwinds, key takeaways and macro

This theme covers companies that make physical things in their own plants, such as metal parts, magnets and molecules, and use their own software and AI to make them faster and with fewer skilled people than older competitors. They earn money on what they ship. We call them vertically integrated because one company owns the whole job, from taking the order to shipping the product. The takeaways below cover why demand is rising, what makes these businesses hard to build, and where the money ends up.

1. **Some tailwinds behind US production are lasting and others can reverse. The lasting ones are what we should build a thesis on.**
   - **Lasting:** China's control of mineral refining and magnets, the shortage of skilled factory workers, and the years it takes a big buyer to approve a new supplier. None of these goes away with an election or a budget cycle.
   - **Temporary:** tariffs, the current surge in munitions spending and interest rates. Tariffs are set by proclamation and can change overnight. Munitions money depends on bills Congress has not passed. The Fed raised rates to 3.75-4.00% in September, and spending on new factory construction was down 21.9% year on year by May 2026 (Federal Reserve, ENR).
   - **Implications:** we should favor companies that still work if tariffs fall and the munitions surge slows. That means owning approved capacity that buyers need for a decade, and avoiding businesses whose margin exists only because of a tariff.
2. **China's hold on minerals and magnets is the most lasting limit in the theme.**
   - China leads refining in 19 of 20 strategic minerals, with about 70% average share (IEA). Refining is the step that turns ore into pure metal. China also makes about 94% of the world's rare-earth magnets (Mining.com). These magnets go into drones, missiles, motors and robots.
   - The US is fully import-reliant for 16 critical minerals (USGS). China has already used export controls on these materials, and its current pause on those controls ends on November 10 and November 27, 2026.
   - This is why buyers and the government will pay for a non-Chinese supply even when it costs more. It also means the plants that fix this problem are expensive and slow to build.
3. **Defense demand and government money are real, but much of the money is conditional and slow.**
   - The FY27 budget request asks for $49.5B for 10 key munitions, about 8.5 times the past decade's average. About 82% of that depends on a reconciliation bill Congress has not passed (CNAS). The Navy is putting $900M into plants that make submarine parts (Breaking Defense).
   - The FY26 defense bill lets the Pentagon buy munitions over several years and commit to purchases of up to 10 years, with advance payments. That is new and helpful, though each year still needs Congress to approve the money (FY26 NDAA).
   - Government now acts as lender and investor. The Pentagon's Office of Strategic Capital (OSC), which lends to companies in key supply chains, has made a $620M loan to one magnet maker and a $500M conditional loan to a rare-earth metals maker, and about $20B is requested for FY27 (Mintz). The Department of Energy's loan office has over $289B of lending authority (DOE). Governments worldwide put about $65B into minerals in 2025 (IEA). These loans take a long time to close and come with conditions.
4. **The labor gap is lasting, and it is driving owners of small shops to sell.**
   - US manufacturers need to fill about 3.8M jobs by 2033, and about 1.9M of them may go unfilled (Deloitte).
   - Of 17,156 US machine shops, 83% have fewer than 20 employees (Census). Many owners are near retirement with no one to take over. That creates a supply of certified shops for sale at reasonable prices.

- Fewer skilled people means a plant that needs fewer of them can grow where others cannot. That is the opening for software and automation.

5. **Capex before contracts is the central risk. A factory has to spend its money before it wins the orders.**
   - Capex means capital spending on buildings, machines and certifications. A big defense contractor, often called a prime, won't approve a new supplier or sign a volume contract until the plant exists and is certified. Approval takes about nine months to two years: one prime told us 9-12 months in an expert call, and another said one to two years.
   - The numbers are large. One precision-parts maker committed about $200M to a single Arizona plant (Plant Services), and its CEO says capex runs about two years ahead of revenue (claim). Process plants need about 80-85% use of capacity to make money (FTI Consulting). A hardware company at Series B often has about $10M of revenue and break-even gross margins (Lowercarbon).
   - This hurts venture returns. Equity spent on a building earns building-like returns and dilutes founders before the first big order arrives.
   - The winners shift who pays for the build. Options include government loans and grants, a buyer that agrees to purchase output before the plant is built (the Pentagon agreed to buy all of one magnet plant's output for 10 years, with a price floor), a customer that hosts the factory inside its own plant (one 7-year deal), bank debt once revenue exists, leased buildings, franchising, and buying plants that are already certified. One roll-up founder said they buy shops at about seven times annual earnings (claim).
6. **Buyers pay for speed and spare capacity. Few pay for a lower price, so the margin goes to whoever owns the output and holds the approvals.**
   - "Cost is secondary. Cost is driven usually by schedule," a director at a large defense contractor said in an expert call. An engineer there would pay 20-40% more to avoid a delay. The premium is highest on prototypes, spare parts and urgent orders, and it fades at volume.
   - Owning the plant pays. One listed parts maker earns a 49% gross margin in its own plants and 31% on work it farms out to partner shops (company filings). Approved incumbents earn well too: a large engine-parts unit runs a 37.7% EBITDA margin (earnings before interest, taxes, depreciation and amortization). Aerospace programs can run about 50 years, so an approval keeps paying.
   - Selling software or machines to factories and mines has been hard. One 3D-printer maker fell from a $2.1B value to a $42.5M sale (3D Printing Journal). An AI-run machining startup in Germany went insolvent in August 2026. Buyers cap each supplier at about a quarter to a third of their work, so even a winner grows one share at a time.
7. **AI's edge today is in the work around the machine and in processes that run nonstop. Nobody has shown yet that it makes each unit cheaper as volume grows.**
   - Around the machine means quoting, planning, writing the instructions a machine follows (CAM programming), inspection and paperwork. One startup claims it cut assembly-instruction writing from about 10 hours to 15 minutes (claim). A former employee of one parts maker said its time from order to first part fell about 50% over two years.
   - The machines themselves are underused. In a survey of more than 100 plants, cutting machines ran about 34% of the time against a 70% target. European process plants run at about 74% of capacity against the 80-85% they need (FTI Consulting). Better planning and control software closes that gap.
   - In a plant that runs nonstop, like a chemical or metals line, every extra point of yield (good output per unit of input) is close to pure profit. A head of strategy at a large aircraft maker put net savings from automation, after paying for the machines, at 20-30% at best. Labor is under 10% of the cost of goods in a new automated plant (claim), so the case rests on speed and yield more than headcount.
8. **The key risk is that incumbents automate and buy faster than new companies can grow.**

- The primes decide who is approved and how much work each supplier gets. The largest aerospace parts supplier saw its share of one buyer's machining fall from about 50% to 30%, and it is now automating (expert call).
- Big companies buy instead of build. Miners back startups through their venture arms, one paid $6.7B for a lithium producer, and a defense contractor's venture arm led a round in an AI materials startup. Drug companies are building their own AI labs, one with plans for up to $1B.
- There is still room. The Pentagon relies on more than 200,000 suppliers and has country-of-origin data for fewer than 10% of the lower-tier suppliers on the F-35 (GAO). Most of that base is small, old and short of people, which is where new companies can win.

2. Interesting categories

This section lists every category in our primer where our view is positive or possibly positive, grouped the same way the primer is. Where the economics differ inside a category, we split it into smaller pieces. Each piece ends with the companies the primer names there, so we can decide together which categories to keep.

A. Raw Materials

Raw materials has two stages: finding and mining ore, then refining it into pure metals and magnets. China's hold on refining makes this the most lasting tailwind in the theme, but these are also the most capital-heavy businesses. Our primer view is selective: attractive in refining when someone else pays for the plant, and early in exploration.

Mining: AI exploration that sells deposits

- **Why it's interesting:** Finding a new deposit is slow and mostly fails. One explorer puts industry success at about 0.5% per project (claim). US mines take about 19 years from discovery to production. AI can narrow where to drill, and the big miners are paying for that through venture arms and accelerators.
- **How the business works:** The explorer owns the mining claims and drill rigs, proves out a deposit and sells it to a miner, for example for about 5% of the metal in the ground plus a royalty (claim). This is a hybrid model: it owns the find but not the mine. Costs are drilling and geologists, far less than a mine, but revenue comes in lumps years apart. Software sold to miners on its own rarely sticks; one explorer ran about 80 pilots that did not turn into repeat business.

Example companies: Earth AI, KoBold Metals, Terra AI, GeologicAI, Ideon

Mining: restarting and running mines with automation

- **Why it's interesting:** Data centers and the grid need far more copper; one AI campus can need up to 50,000 tons. Restarting an old, permitted mine skips much of the permit wait, and automation cuts the number of people needed.
- **How the business works:** The company owns the mine and plant and sells metal, such as copper cathode, at market prices. One startup restarted a Utah copper mine in May 2026 and says it runs about half autonomously (claim). It is fully owned and very capital-heavy: about 5 years to build and 3 to 5 to ramp after permits (claim). Margins swing with commodity prices, and no startup in this group has public buyer contracts.

Example companies: Mariana Minerals, BHP, Rio Tinto, Caterpillar

Refining: plants backed by government debt and signed buyers

- **Why it's interesting:** Chinese refineries take about 6 months to commission against 2 to 4 years in the West. Buyers in defense and autos need non-Chinese metal and magnets, and the government is lending and signing long purchase deals to get them built.

- **How the business works:** The company buys concentrate or scrap and sells refined oxides, metals or magnets under long contracts. It owns the plant, but government loans, grants and a signed buyer cover most of the build. Examples of the money: a $620M loan plus $50M of equity to one magnet maker, a $500M conditional loan to a rare-earth metals maker, and for the largest incumbent a 10-year price floor and a promise that all output of its next magnet plant will be bought. A federal tax credit covers 10% of production cost until it phases out in 2031-34. Even so, that incumbent lost $85.9M on $224.4M of revenue in 2025.

Example companies: Vulcan Elements, Phoenix Tailings, ReElement, Ucore, Energy Fuels, MP Materials

Refining: plants funded mainly with equity

- **Why it's interesting:** The same China problem applies, and tariffs help: one large steelmaker's margin rose from 14% to 20%, and steel mill product prices are up about 86% since February 2020. Some startups say software can halve a 5 to 10 year build (claim).
- **How the business works:** The company owns the refinery and sells metal at prices set globally or in China, so without a price floor it is a price-taker. Lithium fell 24% and nickel 11% in 2025. A refinery is one giant bet, and one founder called custom lines "unfinanceable" (claim). Equity pays for most of the build, which is the capex-before-contracts problem at its worst.

Example companies: Mariana Minerals, Hertha, Cyclic, Mass Magnetics, Duranium, Boston Metal, Electra, Ceibo, Nucor, Carpenter Technology

B. Manufacturing

Manufacturing is the most commercial part of the theme. Most companies here already have revenue. It covers metal supply, fabrication, precision and defense hardware, and buying certified shops. Our primer view is attractive for fabrication, selective for precision and defense hardware, and low conviction for metal supply.

Metal supply: distributors that cut metal to size

- **Why it's interesting:** Tariffs and buy-American rules help US metal, and job shops wait days for a quote from a broker. An online distributor can quote in minutes (claim).
- **How the business works:** The distributor buys metal from mills, holds it in stock, cuts it to size and sells it to shops and primes. It earns the spread between buying and selling plus cutting fees. Plants are cheap, but a lot of cash sits in inventory. Instant quoting is easy with a big safety margin; the hard part is beating a human estimator (claim). Incumbents are gaining share without AI, so this is our lowest-conviction manufacturing category.

Example companies: Reliance, Ryerson

Fabrication: online custom parts from owned plants

- **Why it's interesting:** Hardware companies of every kind need custom metal parts fast. One online fabricator handles about 32,000 orders a month and is growing about 100% a year, with over 300,000 customers. It says it has been profitable for eight years (claim). Its lasers run 86% of the time thanks to scheduling software.
- **How the business works:** Customers upload a design, get an instant price and receive parts in days. The company owns its lasers, press brakes and finishing lines, which cost a fraction of a precision plant, and earns a margin on each order. It wins on speed: its founder says he beats China on price on only about 20-25% of his work (claim). The risk is that customers outgrow it and buy their own machines.

Example companies: SendCutSend, 1872

Fabrication: partner networks

- **Why it's interesting:** A network can offer the same instant quotes without building plants, which makes it the lightest-capital way into fabrication.

- **How the business works:** The company takes the order and routes it to partner shops, keeping a cut. This is hybrid. It needs little capital but keeps less of each sale: one listed parts maker earns a 49% gross margin in its own plants and 31% on work it sends to partners (company filings). One startup sends about 90% of its volume to partner shops. Quality control across shops is the main risk.

Example companies: Jeevy, Protolabs

Fabrication: robotic sheet forming

- **Why it's interesting:** Pairs of robot arms shape sheet metal into airframe and car panels without the costly dies a stamping press needs. A new die takes months and a lot of money, so this suits defense programs whose designs change often.
- **How the business works:** The company sells formed parts and paid projects. It runs its own plants and sometimes places robot cells at customer sites, which makes it hybrid. Each part takes minutes to hours against seconds on a press (MMS), so it wins on short runs. One company says it won't start a project unless the customer commits money, which limits capex before contracts.

Example companies: Machina Labs

Fabrication: wire harnesses and electronics assembly

- **Why it's interesting:** A wire harness is the bundle of wires and connectors that carries power and signals through an aircraft, satellite or missile. Most are still built by hand, and a late one holds up the whole system. One maker claims it cut lead time from six months to two weeks (claim).
- **How the business works:** The company sells finished harnesses or assembled boards per program. Software turns the design into a digital model and step-by-step instructions, so more units pass inspection the first time and less experienced workers can do the job. Plants are owned but light, mostly benches, testers and people. One maker leased a second plant of about 80,000 square feet and opened it within months. Growth so far comes from existing customers.

Example companies: Senra Systems, CircuitHub

Precision and defense hardware: owned factories for machined, cast and composite parts

- **Why it's interesting:** Precision parts are metal or composite parts made to very tight limits for missiles, jets, satellites and submarines. The shops that make them are small and slow, and primes, which earn 7-12% on cost-plus contracts, outsource much of this work. A supplier that delivers fast earns a premium and, once approved, keeps the work for programs that run decades.
- **How the business works:** The company runs its own plants, sells parts under purchase orders or multi-year contracts, and keeps the whole margin and all the process data. It also pays the whole build before approval: one parts maker committed about $200M to a single plant, and a former employee said a six-machine first plant cost "well north of $10 million." Debt and Navy co-funding help once revenue exists. Approval takes AS9100 certification, the aerospace quality standard, which typically takes 6-12 months, then each part's own qualification at the buyer.

Example companies: Hadrian, Fabri, Layup, Hybron, Furientis, Helicon, Westmag, Anduril, Precision Castparts, Howmet

Precision and defense hardware: 3D-printed parts from owned printers

- **Why it's interesting:** Metal 3D printing can make shapes machining can't, with no tooling, and it fits parts that carry heat or fluid and printed missile structures. Printers that check each layer with cameras build the best process data in the theme.
- **How the business works:** The company builds or buys its own printers, keeps them and sells parts per unit. Selling printers to others has gone badly: one printer maker fell from a $2.1B value to a $42.5M sale (3D Printing Journal). Owned printer plants are large bets; one 430,000-square-foot factory is about 80% sized for defense work still waiting on budgets. One drone founder said 3D printing is "way high on the hype curve" (claim).

Example companies: Divergent, Freeform, Rangeview

Precision and defense hardware: tooling and molded parts

- **Why it's interesting:** Almost every plastic part is made in a steel mold, and designing and building that mold takes skilled toolmakers many weeks. Simulation and machine learning can shorten that.
- **How the business works:** The company designs and builds the mold in its own shop, then runs it in its own presses and sells finished parts. Owning both steps keeps the margin on the long run of parts after the mold is done. One company claims "mid eight figure" part contracts (claim). Plants need a toolroom and presses, smaller than a precision machining plant.

Example companies: Atomic Industries

Precision and defense hardware: franchised or hosted-cell models

- **Why it's interesting:** This is the lightest-capital way into precision parts. It answers the capex-before-contracts problem by putting someone else's money into the machines.
- **How the business works:** In a franchise, franchisees buy the machines while the company supplies software, orders and quality rules and keeps a share. One network claims 14 factories, 4 owned and 10 franchised, and a lead-time cut from about 10 weeks to under 14 days for one customer (claims). In a hosted cell, the company runs its machines inside a buyer's plant; one 7-year deal places cells inside a prime's missile plant. Both keep less of each sale but scale faster. The question is whether primes accept the same quality across many sites.

Example companies: Isembard, Hadrian

Buying and automating certified shops

- **Why it's interesting:** Thousands of small-shop owners are retiring. Their plants already hold certifications and approvals, often on single-source parts, so buying one skips most of the approval wait.
- **How the business works:** The company buys a profitable certified shop, keeps its customers and adds software and automation to raise output and margin. It is owned by acquisition: the machines are already paid for, so build risk is low, and the risk moves to the price, the debt and keeping key engineers. Incumbent roll-ups run about a 30% gross margin and buy at about seven times annual earnings (claim). Some defense contracts can be voided when a supplier is sold, which needs checking deal by deal.

Example companies: Amca, Re:Build

C. Specialty Chemicals

We limit chemicals to automated chemistry labs that make new molecules for drug programs. Drug companies send most of this work offshore, largely to China, and the two biggest offshore providers are worth close to $100B combined (Lightspeed). BIOSECURE, a law passed in December 2025, limits use of certain Chinese biotech suppliers, and 79% of 124 biotechs surveyed had a China-based contract manufacturer (BIO). Our view: early, and they need paying customers.

Robotic chemistry platforms

- **Why it's interesting:** These labs turn molecule designs into code that robots run, and log every result, including failures most labs throw away. That data makes the next prediction better.
- **How the business works:** The lab sells discovery chemistry and co-discovery projects to drug companies and research funders, at milligram-to-gram scale for now (claim). It owns the lab, the software and the reaction data. Labs cost tens of millions; one opened a £12M first lab and is part-funding an £89.9M expansion with about £22M of UK and Scottish grants. One claims to work with eight of the top 20 drug companies (claim).

Example companies: Chemify

US chemistry service labs for drug discovery

- **Why it's interesting:** This is a direct US replacement for the offshore labs that make the molecules drug teams test, and BIOSECURE pushes buyers toward it.
- **How the business works:** The lab sells chemistry work to biotech and pharma, combining chemists, AI and robots. It does not develop its own drugs and may later sell predictions. US chemists cost several times Chinese rates, so the business works only if automation brings unit cost down. Capex is lab buildout plus a large team. The leading startup launched in October 2026 with design partners and no public customers.

Example companies: Warp Sciences

D. Advanced Materials

Advanced materials are new materials such as battery cathodes, cooling fluids and alloys. AI can find candidates in weeks, but making them at scale and getting customers to qualify them takes one to five years. Our primer view is selective for producers and early for discovery labs.

Materials producers

- **Why it's interesting:** China makes about 85% of battery cathode material and over 98% of the cheaper LFP type. Data centers used 415 TWh of power in 2024, heading to about 945 TWh by 2030, and cooling takes 7% to over 30% of it. Liquid cooling is expected on 53% of AI chips in 2026. Both create buyers for a non-Chinese or new material.
- **How the business works:** The producer designs the material and sells it by the kilogram or ton under long qualification and supply deals. Some own the plant; others use a contract maker, which is a hybrid with far less capex. Owned cathode plants cost hundreds of millions and compete on Chinese prices. Even an established incumbent's battery-materials unit lost money on €436M of revenue.

Example companies: Chemix, Albemarle, Umicore, Corning

Materials discovery labs

- **Why it's interesting:** AI plus robotic labs can test far more candidates than people can. One lab made about 1,200 alloys in five or six months (claim), and an independent university test found one of its alloys beat a standard aerospace alloy.
- **How the business works:** Labs earn from paid R&D for partners, licensing and royalties, and government research awards, while a partner makes and sells the material. They don't own the output, so their share is a fee or royalty. Government awards run around $1M. Valuations are set on talent, from a $300M seed to one reported at about $7.5B. One investor says materials has had "no AlphaFold moment."

Example companies: Radical AI, CuspAI, Periodic Labs, Lila Sciences, Aionics, Mattiq, Dunia Innovations, Entalpic, Discovered Materials, Kebotix

Materials R&D software, licensed

- **Why it's interesting:** Chemical and materials companies want AI tools for their own R&D teams, and some of the largest are already paying.
- **How the business works:** The vendor sells a data platform and AI tools to corporate R&D teams on multi-year licenses; one claims 10 such customers (claim). It is recurring software revenue with no plant, but the customer keeps the product margin.

Example companies: Citrine Informatics, Uncountable, Albert Invent, Microsoft Discovery

E. Enabling Software & Robotics

These companies sell tools to factories and plants instead of making the product. The primer tracks them outside the core, because selling to plants has been hard and incumbents own the installed base. The split that matters is how they get paid: on results, by license, or by renting robots.

Plant software paid on results

- **Why it's interesting:** Process plants run below the capacity they need, about 74% in Europe against 80-85% (FTI Consulting). In a sold-out plant each extra point of yield is close to pure profit. This is the closest fit for an enterprise software fund.
- **How the business works:** The software connects to a plant's existing controls, adds cheap sensors, builds a live model and adjusts setpoints itself through a safety layer. A setpoint is a target such as a temperature or flow rate. One startup says it is paid on the production it adds (claim). There is no factory; the cost is engineering and long deployments. The hard parts are agreeing on a baseline and getting the plant's safety team to approve software that makes changes on its own.

Example companies: Lumaril

Licensed factory software

- **Why it's interesting:** AI speeds the office work around the machine. One tool claims to cut assembly-instruction writing from about 10 hours to 15 minutes (claim), and others write machining programs or inspect welds.
- **How the business works:** The vendor sells software by subscription to factories. There is no plant, but the customer owns the output and the data. Most startups here have a first customer and little else, and big automation vendors bundle software with hardware at high margins; one runs a 49.5% gross margin. These tools may help old shops catch up as much as new ones.

Example companies: Arda, CloudNC, AI Robotics, Siemens, Rockwell Automation, Honeywell, AspenTech, Emerson

Robots as a service

- **Why it's interesting:** The US installed 38,428 industrial robots in 2025 against 354,226 in China, and has 307 robots per 10,000 workers against 1,220 in South Korea. The labor gap makes renting automation attractive to small plants.
- **How the business works:** The vendor owns turnkey robot cells, places them in the customer's plant and charges for use. It is hybrid: the vendor pays for the robots and the customer hosts them. One company found selling autonomy software alone "doesn't really work" and switched to renting cells, and it claims one customer went from about 300 to 1,700 units a day on the same robots (claim).

Example companies: GrayMatter Robotics

Robot AI sold to plants

- **Why it's interesting:** Cheaper robots and better models could automate tasks like picking and fastening that still need people.
- **How the business works:** The company trains robot models and sells robots or software to plants. Robot-learning success is 59-66% on short tasks against the 99.9% plus a factory needs. One company in this group has raised over $1B with no public product and relies on a single partner that owns about a third of it.

Example companies: Mind Robotics, Bright Machines, Rapta, Allus, Faraday